Samuel Dawit Assefa

Senior at KAIST (7th Semester)
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github.com/sammythedude Nationality: **Ethiopian**

Education

Korea Advanced Institute of Science and Technology (KAIST) **Daejeon, South Korea**
Bachelor of Science in Computer Science **2023 – 2027**
Relevant Courses: **Statistical Machine Learning**, **Artificial Intelligence and Machine Learning (CS570)**, **Introduction to Artificial Intelligence (CS470)**, **Intro to Deep Learning**, **Probabilistic Machine Learning (Individual Study)**

Research Interests

Parameter-efficient LLM alignment; small-model augmentation (distillation, RAG, tool use); hallucination detection and quantitative evaluation; uncertainty and calibration in language models.

Conferences

- **Biophysics Division, 2026 Korean Physical Society (KPS) Meeting** — First-author conference abstract on **AI-based computer vision for live-sperm DNA fragmentation analysis**, applying deep learning to 3D holotomography imaging for single-cell prediction.

Professional Experience

AI Engineer Intern **Dec 2025 – Feb 2026**
PuzzleAI (Mentor: **Harin Jun**, CTO) Daejeon, South Korea

- Designed and implemented **quantitative hallucination metrics** for LLMs in medical-domain settings, achieving $> 90\%$ **accuracy with** $< 2\%$ **variance** against gold labels across evaluation runs.
- Built an **automated benchmarking pipeline** combining LLM-as-judge and rule-based evaluation for scalable validation of hallucination detection metrics.
- Investigated **attention- and logit-based interpretability signals**, measuring correlation with hallucinated outputs across prompts and decoding strategies.

Researcher in Computer Vision & Data Science **July 2025 – Present**
BioMedical Optics Lab, KAIST (Advisor: **YongKeun (Paul) Park**) Daejeon, South Korea

- Developed **label-free computer vision pipelines** on 3D holotomography imaging for single-cell sperm analysis, extracting engineered refractive index (RI) features.
- Finetuned segmentation models and built interpretable machine learning models to work on top (ElasticNet with interactions) achieving **ROC-AUC 0.739** and **PR-AUC 0.648** on curated datasets for DNA fragmentation prediction.
- Conducted translational validation on human sperm datasets, confirming feasibility of **live, label-free single-cell DNA fragmentation assessment**.
- Presented as **first author** at the Biophysics Division, 2026 Korean Physical Society (KPS) Meeting.

Research Intern and System Designer (LLM-based Systems) **Winter 2024 – July 2025**
KAIST Interaction Lab (KIXLAB) (Advisor: **Juho Kim**) Daejeon, South Korea

- Designed and prototyped **retrieval-augmented generation (RAG)** systems for codebase understanding and developer assistance.

- Conducted research on **human–LLM interaction in programming**, analyzing how users explore and reason about large software systems with LLM support.

Individual Study Researcher **Fall 2025**
Brain and Machine Intelligence Lab (KAIST) Daejeon, South Korea

- Conducted an individual research study on hierarchical LLM agent systems for human–AI collaboration.
- Identified common low-level execution failures in grounded LLM agent architectures operating in interactive environments.
- Implemented a prototype tool-calling agent using API-based actions for controlled evaluation and deployment.

Projects

- **Custom Language Model Training:** Curated a small-scale instruction-style dataset and fine-tuned a language model, implementing data preprocessing pipelines and evaluating generation quality under limited data regimes.
- **StyleCLIP Latent Mapper:** Improved model disentanglement by implementing hierarchical latent mappers beyond three levels; trained on self-managed GPU servers.
- **Vision Transformer on MedMNIST:** Implemented a hybrid convolution–ViT architecture combining locality and global context for medical image classification.
- **FCN-based Segmentation on PASCAL VOC:** Implemented an FCN-based segmentation model achieving 89% pixel accuracy and 0.60 mean IoU.
- **Hilbert Curve and Spatial Relationships in CNNs:** Explored space-filling curves to transform convolutional operations into 1D sequences while preserving spatial locality.
- **MNIST-GAN and Mode Collapse Analysis:** Analyzed the impact of loss function choices on GAN stability and mode collapse.

Skills

- **Programming:** Python, C, JavaScript
- **Libraries:** NumPy, pandas, matplotlib, seaborn, scikit-learn, PyTorch, HuggingFace Transformers, React
- **Tools:** Jupyter, Git, Docker
- **Languages:** English (Native), Amharic (Native)

Other Experience

- **Student Assistant,** International Students and Scholars Services (ISSS), KAIST (July 2023 – Present): Supported staff in organizing campus events for international students.
- **Volunteer Mentor,** Hanwha-KAIST Mentorship Program (Apr 2023 – Dec 2023): Led English lessons and mentorship activities for middle school students from underprivileged backgrounds.